

Hanif Ridho Abdillah

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ABOUT ME

A Geophysical Engineering student with a strong interest in subsurface exploration, particularly in seismic data interpretation and Carbon Capture Storage (CCS). I enjoy delving into geophysical data analysis to understand geological structures and assess subsurface resource potential and evaluation. Driven by a passion for learning, perseverance, and a desire to grow, I actively engage in research and organizational activities to broaden my knowledge and gain practical experience in the field of geophysics.

EDUCATION

Sumatra Institute of Technology (ITERA)

- Bachelor's Degree in Geophysical Engineering (2022 – 2026)
- GPA: 3.22 / 4.00
- Active in geophysics-related organizations, research, and competitions.

PUBLICATION

Caprock Integrity Evaluation for Geological CO₂ Storage, Central Sumatra Basin

Final Research Project - 2026 - (DOI: [10.22373/p-ipft.v12i2.34352](https://doi.org/10.22373/p-ipft.v12i2.34352))

- Conducted geophysical and geomechanical analysis of the Telisa Formation as a potential caprock for Carbon Capture and Storage (CCS).
- Performed well log interpretation using gamma ray, density, sonic, and elastic property analysis.
- Evaluated brittleness index, acoustic impedance, Poisson's ratio, Young's modulus, and unconfined compressive strength (UCS).
- Integrated petrophysical, rock physics, and geomechanical approaches to assess caprock sealing integrity and mechanical behavior.
- Developed scientific visualizations, crossplots, and presentation materials for research communication.

Screening Potential CO₂ Storage Reservoir, Central Sumatra Basin

Co-Author Research Project - 2026 - (DOI: [10.22373/p-ipft.v12i2.34353](https://doi.org/10.22373/p-ipft.v12i2.34353))

- Developed geological background and regional geological framework of the Central Sumatra Basin.

- Conducted stratigraphic and depositional environment interpretation of the Menggala Formation.
- Compiled and reviewed geological literature supporting CCS reservoir screening.
- Contributed to manuscript preparation and scientific discussion on reservoir characterization for CO₂ storage.

RESEARCH EXPERIENCE

Potential CCS Development in Igneous Rock Formations in Sumatra, Indonesia

Independent Research Project (2025 -Present)

- Conducted preliminary research on the potential application of Carbon Capture and Storage (CCS) in igneous rock formations in Indonesia, particularly Sumatra.
- Investigated subsurface geological characteristics and their relevance for long-term CO₂ mineralization and storage potential.
- Reviewed regional geological, geophysical, and decarbonization-related studies to support CCS development strategies in Southeast Asia.

INTERNSHIP EXPERIENCE

Geophysical Intern – LEMIGAS (Research and Development Center for Oil and Gas Technology) July 2025 – August 2025

- Interpreted 2D/3D seismic data to identify subsurface geological structures and identify hydrocarbon zone.
- Utilized seismic attributes to enhance the accuracy of structural interpretation.
- Integrated regional geological data and well data to improve interpretation reliability.

ORGANIZATIONAL & COMMITTEE EXPERIENCE

Structural Geology Laboratory Assistant (February 2025 – Present)

- Prepare materials and demonstrate the use of tools such as geological compasses and rock hammers.
- Evaluate student lab results and provide academic and technical feedback to enhance understanding of structural geology concepts.
- Teach geological mapping techniques using topographic maps and stereonet.
- Assist in the processing and interpretation of structural data for tectonic analysis and natural resource exploration.

President, SEG Student Chapter ITERA (December 2024 – December 2025)

- Led and managed the SEG Student Chapter ITERA consisting of 40 active members.

- Develop academic programs such as workshops, training sessions, and geophysical exploration seminars with more than 1000 participants.
- Facilitate networking and mentoring opportunities for geophysical engineering students.

Head of Competition Division, Scientific League, ITERA (*May 2024 – November 2024*)

- Designed the competition framework and mechanisms.
- Coordinated competition staff and collaborated with all involved parties.
- Worked closely with lecturers, judges, and external partners, including academia and industry, to improve competition quality.
- Successfully organized competitions with more than 700 participants.

Research and Professionalism Division Staff, SEG Student Chapter ITERA (*2023 – 2024*)

- Developed and conducted research for monthly geophysical trivia content.
- Assisted in organizing geophysics seminars.

Competition Subdivision Staff, Scientific League, ITERA (*August 2023 – December 2023*)

- Structured competition procedures and coordinated with other divisions.
- Assisted in participant organization and event management.

ACHIEVEMENTS / AWARDS

- Man of The Match Student Competition Himpunan Ahli Geofisika Indonesia Student Chapters Week 2025
- Grand Finalist Student Competition Himpunan Ahli Geofisika Indonesia Student Chapters Week 2025 using Paleoscan Software
- Finalist, Student Creativity Program (PKM) 2025

SKILLS

- **Software Geoscience:** Python, Petrel, HRS, Paleoscan, QGIS, Surfer, Oasis Montaj.
- **Core Competencies:** Well Log Interpretation, Petrophysical Analysis, Seismic Interpretation, Geomechanical Analysis, CCS Screening Evaluation, Structural Mapping, and Reservoir Characterization.
- **Language:** Indonesian (Native), English (Intermediate/Working Proficiency)